

PRAKHAR

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EDUCATION

Graphic Era (Deemed to be University), Dehradun

Expected April 2027

B.Tech, Computer Science & Engineering (AI & ML) | CGPA: 7.78

EXPERIENCE

Associate Software Engineer & AI Lead | Krishnova LLP

Current

- Lead a team building a Retrieval-Augmented Generation (RAG) based AI customer support platform; own the data pipeline design across FastAPI, Docker, ChromaDB (vector database), and HuggingFace models.
- Monitor pipeline performance and output quality using LangSmith tracing — the same monitor-and-catch-issues function that observability tools like Grafana/Datadog serve in production systems.
- Collaborate directly with stakeholders to translate business/support requirements into technical specs and data workflows for the team.

iOS Development Intern | Infosys GEC-2, Mysore (Apple × Infosys Student Developer Program)

- Built Nexus Retail, a native Retail Store Management System (SwiftUI, MVVM) backed by Supabase (Postgres) for real-time data storage, with Razorpay transaction processing and CoreLocation-based geo-analytics.
- Queried and validated relational data in Postgres/Supabase to support store inventory and transaction-level reporting.

RELEVANT PROJECTS

Nova — Voice-Activated Enterprise SQL Agent — SQL, Python, terminal-style query UI [GitHub ↗](#)

- Designed a split-console interface pairing natural-language voice queries with live SQL query traces and a real-time results panel — making raw data instantly explorable for non-technical stakeholders.

URL Shortener — Performance & Security Benchmarking — FastAPI, PostgreSQL, Redis [GitHub ↗](#)

- Built and benchmarked a backend service through three deliberate phases (working → fast → secure); added Redis caching that cut lookup latency by ~99.9% versus a remote Postgres-only baseline.
- Implemented API-key auth, rate limiting, CORS lockdown, and an SSRF guard; documented all performance and security trade-offs.

SHL Assessment Recommendation Agent — FastAPI, Groq (Llama-3.3-70B), BM25 retrieval [GitHub ↗](#)

- Built a retrieval service with synonym expansion and an anti-hallucination validator, iteratively improving Mean Recall@10 to 1.000 across 10 evaluation traces; deployed to HuggingFace Spaces.

OmniTriageEnv — RL Environment for Email Triage — Python, GRPO fine-tuning, Llama-3.2-1B [GitHub ↗](#)

- Built an OpenEnv-compliant reinforcement learning environment for email triage; fine-tuned Llama-3.2-1B using GRPO, achieving a +59% improvement in reward over baseline.
- Deployed the environment and a live demo to HuggingFace Spaces for public evaluation.

MorphUI — AI-Powered Design Color Customization Platform — Vite, Vanilla JS, Figma REST & Plugin APIs [GitHub ↗](#) [Live Demo ↗](#)

- Built a browser tool that parses a Figma file via URL, remixes its entire color system through an AI palette generator (60+ mood/personality presets), and syncs the new theme back into Figma via a companion plugin — no manual hex editing.
- Added real-time WCAG contrast scoring and one-click export to CSS variables, Tailwind config, and JSON tokens; shipped with zero runtime dependencies and deployed live on Vercel and HuggingFace Spaces.

SKILLS

Languages: Python, SQL, Swift, JavaScript

AI/ML & GenAI: RAG pipelines, LLM evaluation & prompting, GRPO fine-tuning, BM25 retrieval, HuggingFace Transformers, LangSmith tracing, Groq/Llama & OpenAI-compatible APIs

Data & Backend: FastAPI, REST APIs, PostgreSQL, Redis (caching, rate limiting), ChromaDB (vector DB), Supabase, API security (SSRF guards, CORS, API-key auth)

Frontend & Design Tooling: Vite, Vanilla JS, Figma REST & Plugin APIs, WCAG accessibility, design-token export (CSS/Tailwind/JSON)

Mobile: SwiftUI, MVVM, Razorpay integration, CoreLocation

Cloud (working knowledge): AWS — Route53, EKS, ALB, RDS, S3 (architecture design & deployment exploration)

Tools & Practices: Docker, Git, data validation & benchmarking, agile collaboration